W4 — Knowledge Ingestion (File Trigger)

Workflow document · Owner: Reem Saleh · Size: 10 nodes (modified: 11) · Synthetic data only · Draft v0.2 (live results 2026-10-18)

What it does

Watches one Google Drive folder of approved reference PDFs, extracts their text, splits and embeds it, and rebuilds the in-memory vector store that W3 searches, keeping the file name, uploader and upload date so answers can cite their source.

How to run it

Add or update a PDF in the folder (polls every minute) or press Execute on Manual Re-ingest. After an n8n restart the store is empty until it runs again.

Prompts used

None. Embeddings: gemini-embedding-2; text splitter overlap 200; metadata fields Reference name, uploader, upload date.

n8n screenshot



W4 canvas after a real run (published, 2026-10-18): 9 files, 528 items.

Test results

Offline = the real code nodes run in Node with synthetic inputs (not an n8n execution). n8n actual = the result of a real run in n8n, filled in during Phase 4.

Case	Kind	Expected	Offline code test	n8n actual
W4-N1	normal	Store rebuilt; a question about its content returns a finding with [Source: Reference name	uploader	creation time].
W4-N2	normal	Store rebuilt with both files; findings can cite either.	-	**Not run on its own; covered indirectly.** Nine PDFs were ingested into 528 chunks in one run; a separate extra file was not added
W4-N3	normal	Same chunks as before; no duplicates.	-	**Live 2026-10-18, 10:38 PM: pass (count only).** 9 files in, 528 items in that run; duplicates inside the store were not checked
W4-E1	edge	No crash; no useful chunks; a question about it returns 'No relevant findings in uploaded references.'	-	**Live 2026-10-18, 11:00 PM: ingestion pass.** Image-only PDF: no crash, store stayed at 528, no notice to the user (I19). The expected sentence belongs to W3, not W1, so the retrieval check was not shown (I21)
W4-U1	unsafe	Modified version: a Telegram notice says the file could not be read; the other PDFs are still ingested. Original export: the run stops.	-	**Live 2026-10-18, 11:11 PM: pass with a gap.** A .docx went to the error branch and Telegram sent the notice; the other 10 files were ingested and the run did not stop. The notice says "File: unknown" (I20)

Known limits

- Any change re-embeds the whole folder into a store that is wiped first.
- The store is in memory and is lost on restart.
- PDF only; the original export has no error branch, the modified version sends a Telegram notice and continues.
- What is in the folder, and whether a toxicologist reviewed it, is not visible in the export.